

Archived STRF-gain local finite-difference pilot

Random full-parameter contexts, small paired gain perturbations, and expected sampled-loss slopes

At the predeclared +/-2% log-gain perturbation, 21/24 contexts were direction-qualified green, 0 were red, and 3 were unresolved. All 24 mean eligibility and finite-difference signs agreed, but only the 21 uncertainty-resolved, stable-sign-across-epsilon comparisons count in the headline.

Direction was much stronger than magnitude calibration. At 2%, Spearman association was **0.75** and cosine alignment was **0.625**. The median resolved FD/eligibility magnitude ratio was **0.16**, while a through-origin fit gave **0.0195**. Thus this pilot supports the archived STRF-gain update's direction in the sampled regime, but not equality to the numerical gradient.

Snapshot	Contexts	Seeds	Trials/context	Gain rows	Forward cell-runs
pre-dPSC local snapshot (Jul 9)	8 LHS batches x 3 cells = 24	123, 456, 789, 1011, 1213, 1415, 1617, 1819	80	center + 12 endpoints + duplicate	2688

Interpretation: each hard-spike trajectory has a discontinuous sampled loss. The numerical quantity here is a paired Monte Carlo central secant of expected sampled PSTH SSE at finite epsilon. It approaches an ordinary derivative only if that expectation is differentiable and the estimates converge as epsilon shrinks.

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What was tested

For each cell-specific Latin-hypercube parameter context, all 11 nuisance parameters were held fixed and STRF gain was evaluated at g_0 and at symmetric log-space probes:

$g_+ = g_0 \exp(h), \quad g_- = g_0 \exp(-h),$ where $h = 0.0025, 0.005, 0.01, 0.02, 0.05, \text{ or } 0.10$.

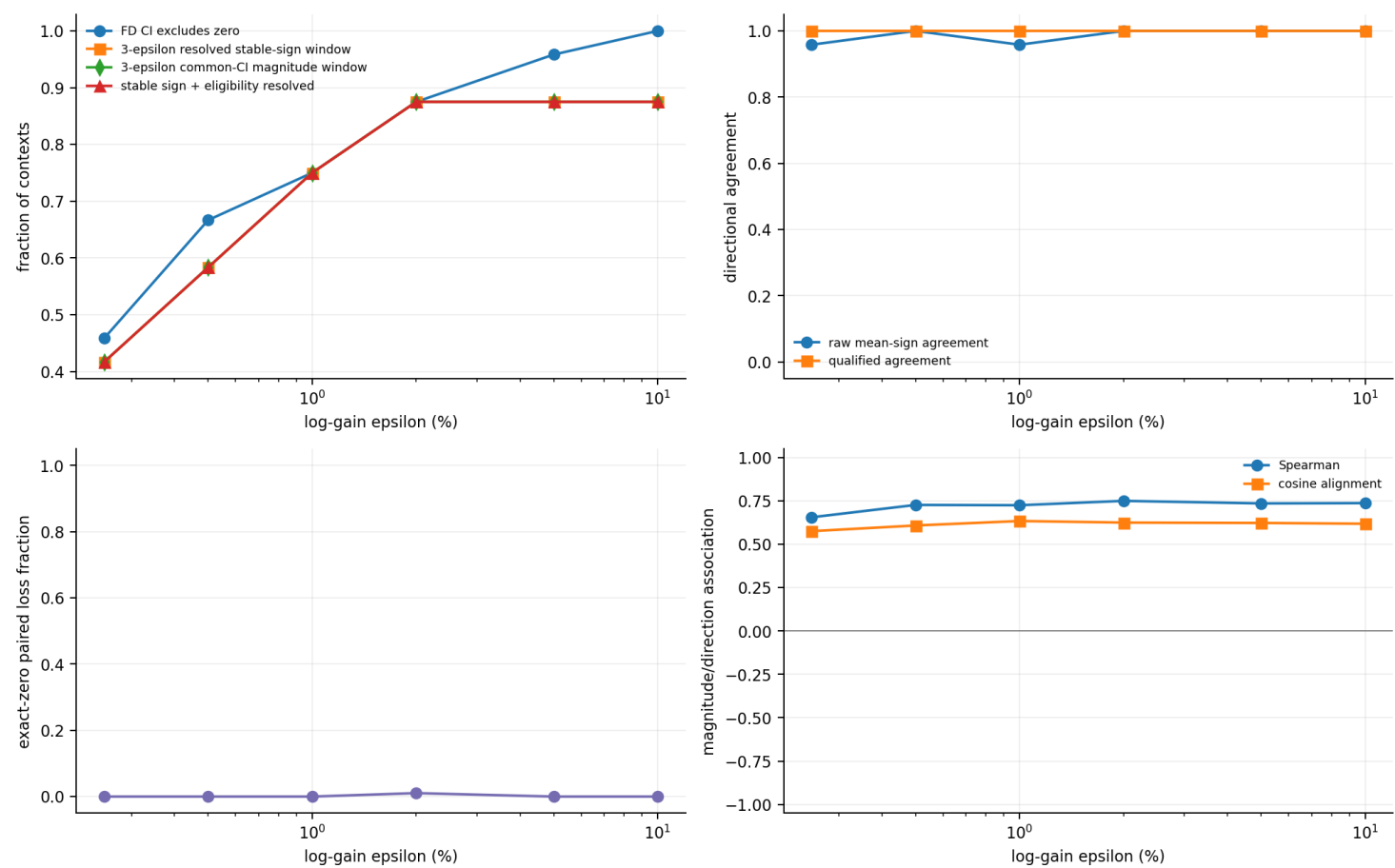
For seed s , the paired numerical log-gain slope is $D(s,h) = [L(g_+;s) - L(g_-;s)] / \log(g_+/g_-)$. The archived center eligibility is expressed in the same coordinate as $A(s) = g_0 \times dL_{\text{elig}}/dg$. Means and nominal 95% t intervals are calculated across eight seed-level paired estimates; each seed contains ten model trials.

Design element	Implementation
Parameter contexts	Separate scrambled Latin-hypercube design for cells 1, 2, and 7 over archived initializer ranges
Gain boundary guard	Gain centers lower-truncated at $0.001 \times \exp(0.10)$, so every -10% log probe remains ≥ 0.001
Paired randomness	Onset, offset, and output-gate torch draws shared only within a center's 14 rows
Independent stochastic replication	Eight sequential seeds; 10 trials per seed
Spontaneous input	Archived numpy spontaneous stream has no batch axis and is shared across centers within each seed
CRN integrity control	A duplicate center row must match loss, eligibility, spike count, and firing rate bit-for-bit
Direction qualification	Eligibility interval excludes zero and FD belongs to a resolved same-sign run across 3 adjacent epsilons
Magnitude stability	Reported separately when 3 adjacent resolved FD intervals have a common intersection
Primary analysis	$\pm 2\%$ log gain, predeclared before final aggregation; all other epsilons are sensitivity checks

Green/red is an exploratory directional label, not a proof of exact differentiation. The t intervals are nominal, the hard-event differences need not be Gaussian, and no multiple-testing correction is applied.

Epsilon sensitivity

Epsilon sensitivity of the STRF-gain finite-difference test



h	FD resolved	Stable sign	FD magnitude-stable	Green	Red	Gray	Raw sign	Spearman	FD/elig median
0.25%	11/24	10/24	10/24	10	0	14	95.8%	0.656	0.214
0.5%	16/24	14/24	14/24	14	0	10	100.0%	0.727	0.0923
1%	18/24	18/24	18/24	18	0	6	95.8%	0.725	0.137
2%	21/24	21/24	21/24	21	0	3	100.0%	0.75	0.16
5%	23/24	21/24	21/24	21	0	3	100.0%	0.736	0.144
10%	24/24	21/24	21/24	21	0	3	100.0%	0.737	0.141

Predeclared 2% result by cell

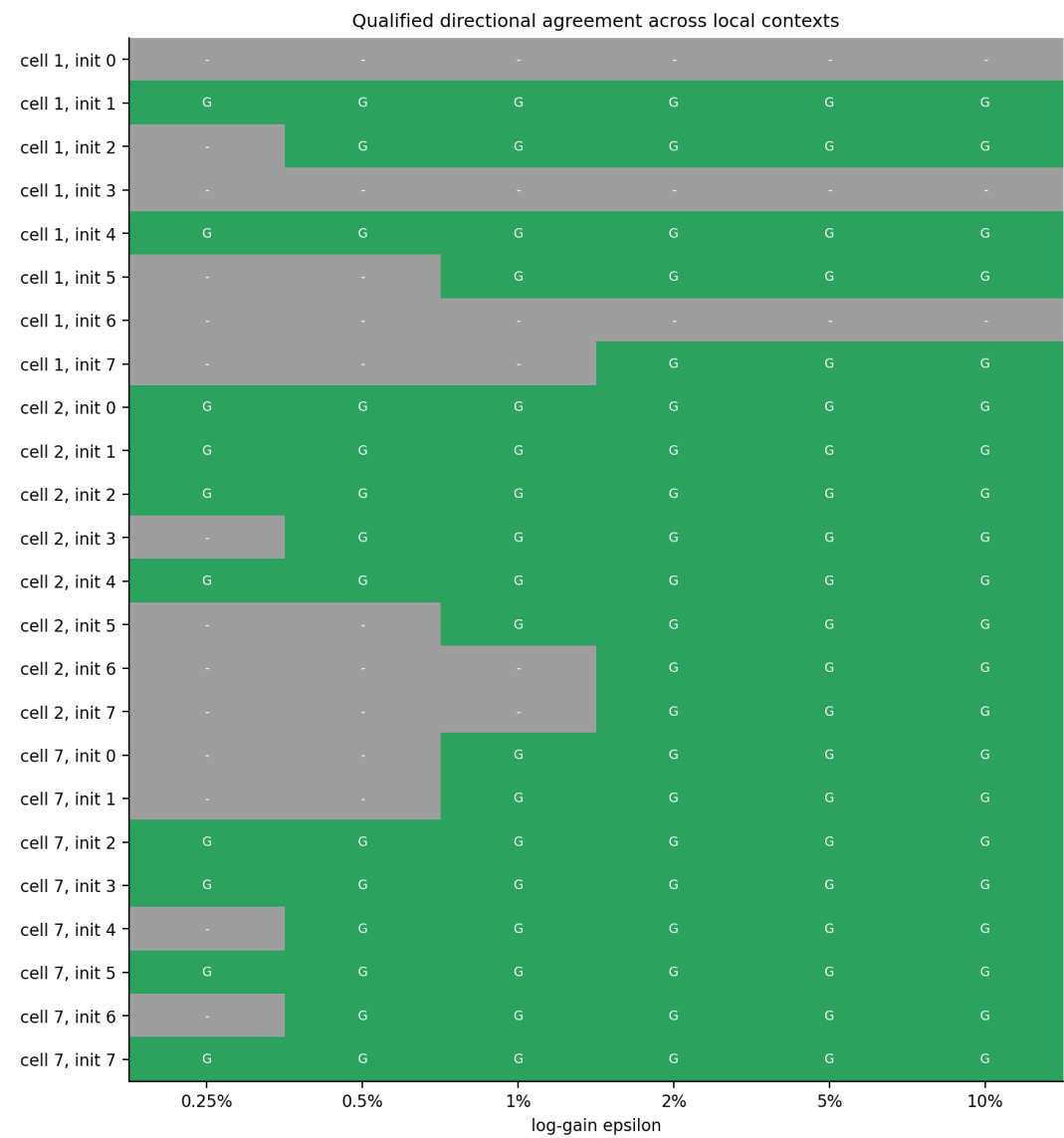
Cell	Contexts	Green	Red	Gray	Median FD/elig	FR min	FR median	FR max
1	8	5	0	3	0.105	37.2 Hz	93.0 Hz	252.9 Hz
2	8	8	0	0	0.375	23.9 Hz	86.7 Hz	304.6 Hz
7	8	8	0	0	0.172	26.5 Hz	99.2 Hz	242.9 Hz

Interpretation

Cells 2 and 7 resolved all eight parameter contexts as green at 2%. Cell 1 resolved five as green and left three gray; none were direction-qualified red. The sole negative-gradient context had a negative finite-difference slope as well. This is substantially more informative than the broad sweep because each comparison is local and paired.

The random initializer produced high firing rates in this pilot (about 24-305 Hz across centers). These are legitimate tests of the original initialization distribution, but they are not a representative sample of fitted or biologically plausible regimes. A trained-state stratum remains necessary before generalizing the result to final inverse-model parameters.

Directional agreement map

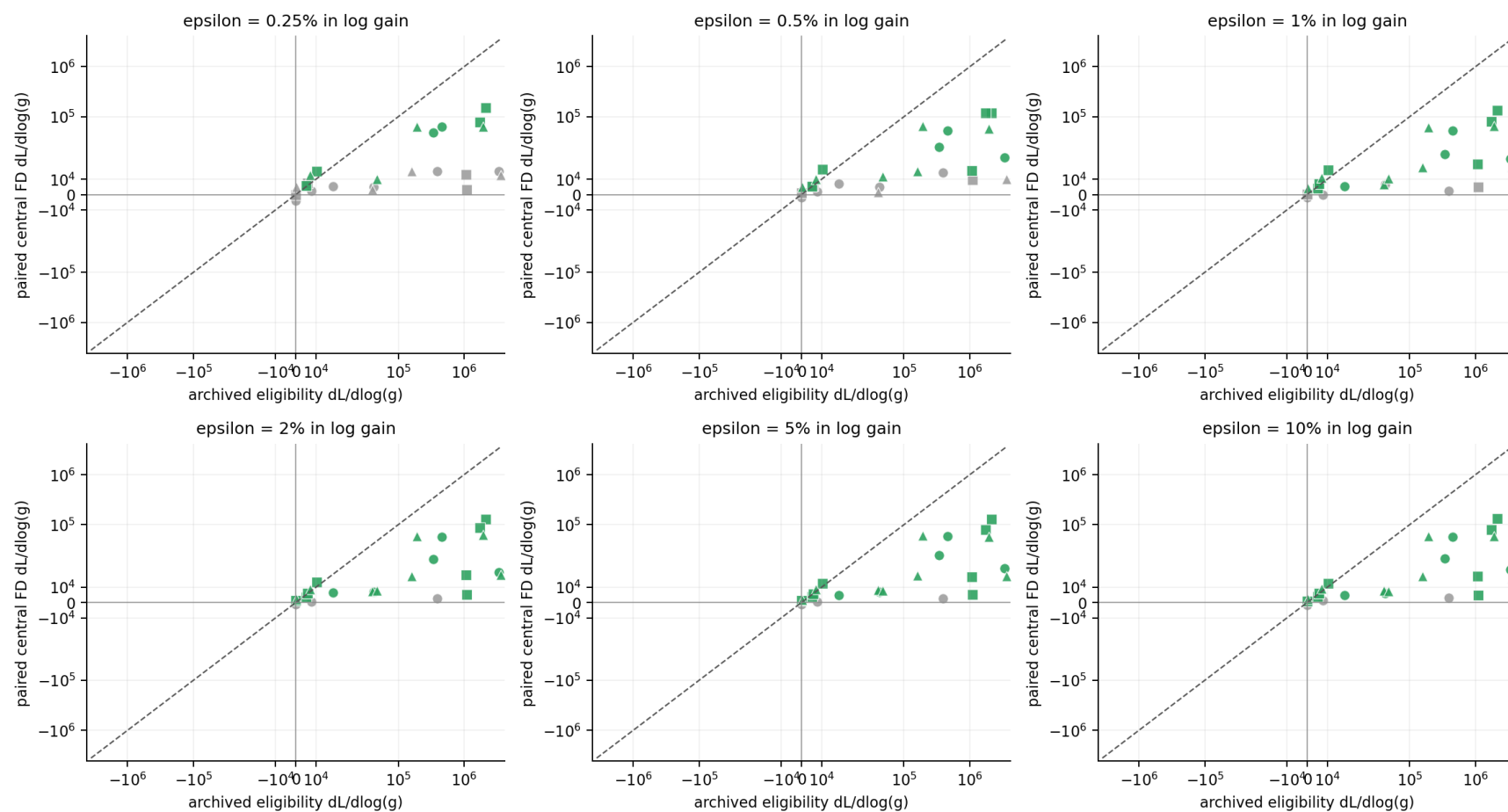


Green requires an uncertainty-resolved archived eligibility and a numerical sign belonging to a three-epsilon resolved same-sign window. Gray means unresolved; it is not counted as incorrect. No red context was observed.

Eligibility versus finite-difference slope

STRF-gain eligibility versus local expected-loss slope

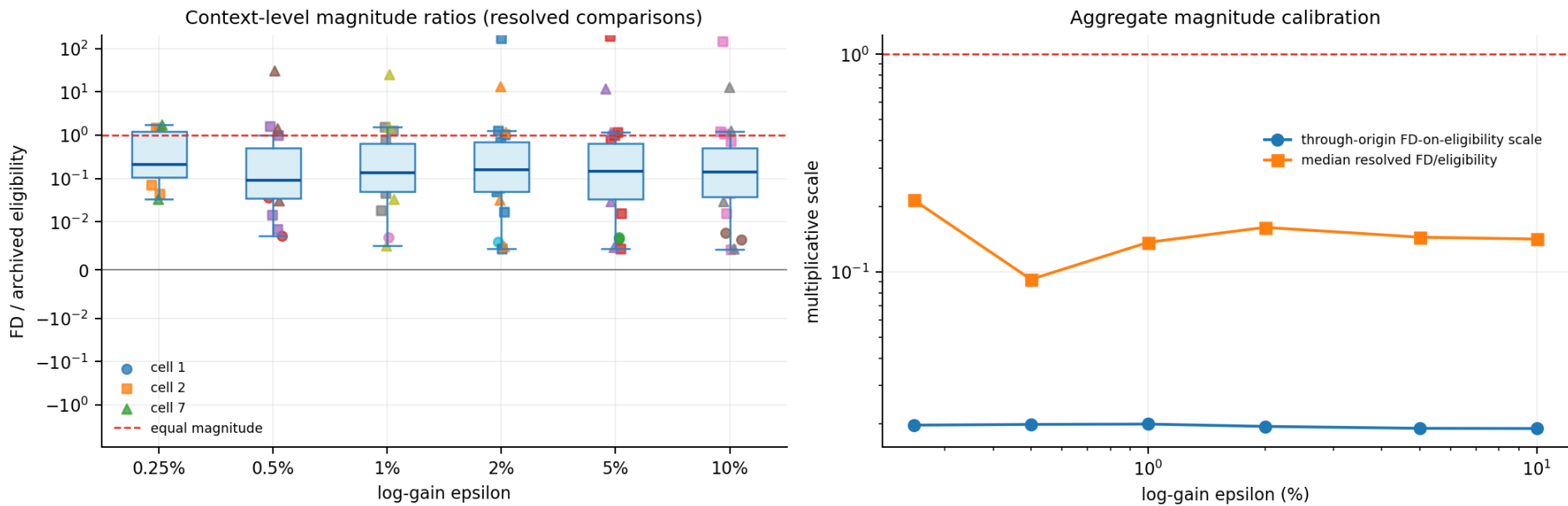
● cell 1 gray ● cell 1 green ■ cell 2 gray ■ cell 2 green ▲ cell 7 gray ▲ cell 7 green



The panels retain all 24 contexts. The identity line tests magnitude as well as direction. Most resolved points lie in the same sign quadrant but well below the identity line, demonstrating directional agreement without magnitude equality.

Magnitude calibration

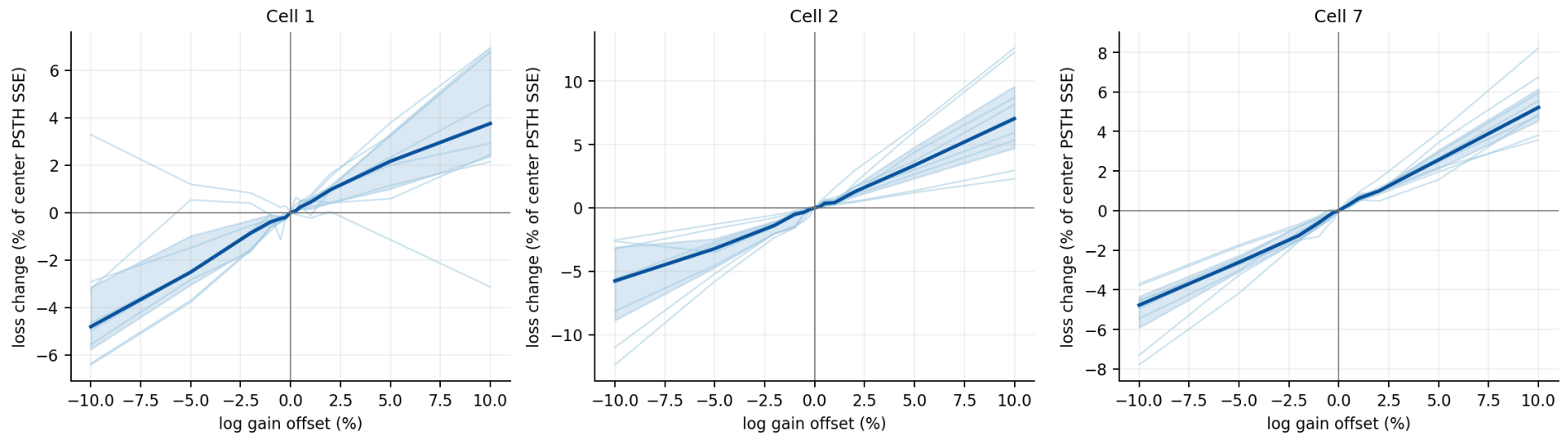
Magnitude agreement is separate from directional agreement



The context-level ratio is heterogeneous. At the primary 2% scale, the median resolved FD/eligibility ratio is about 0.16, but the through-origin scale is about 0.019 because the largest archived eligibility values dominate that fit. Neither statistic is close to one.

Ensemble of conditional local landscapes

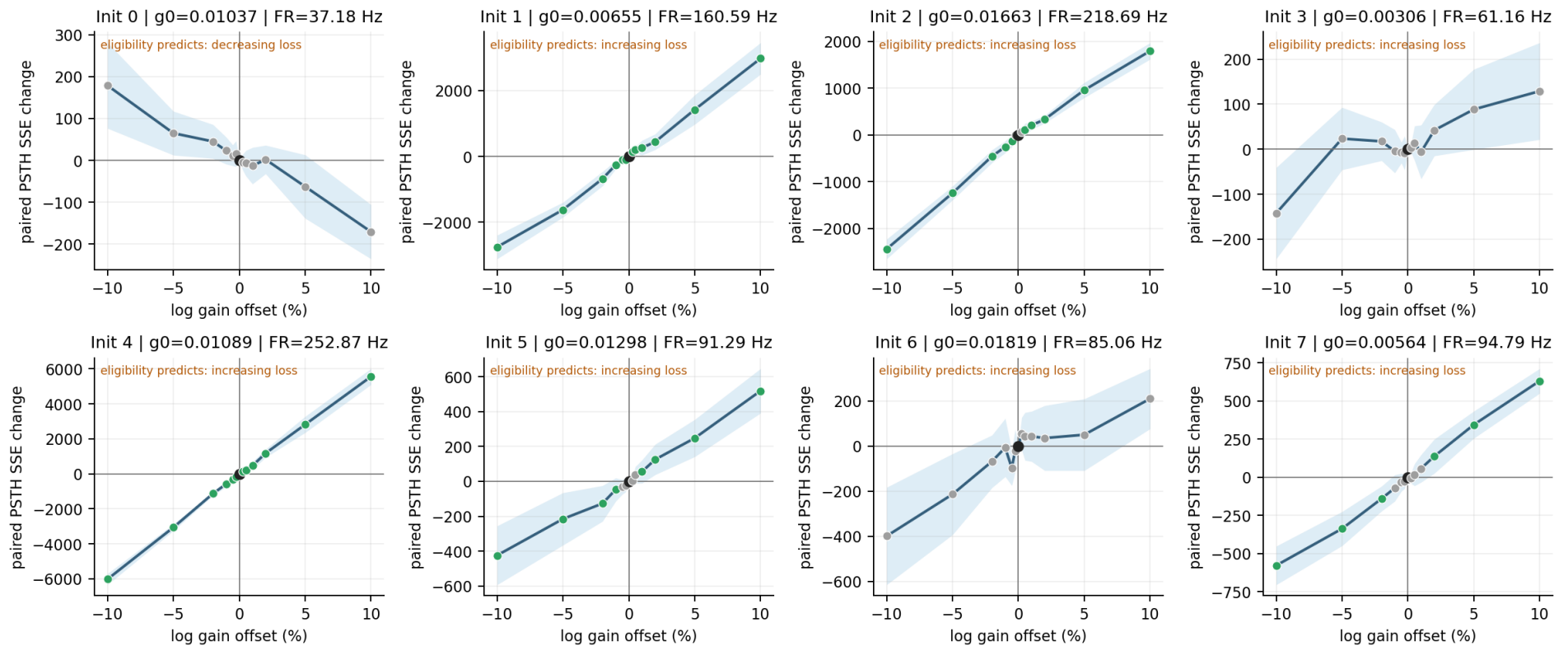
Ensemble of conditional local landscapes (loss change relative to each context's center loss)



Each curve is one conditional one-dimensional slice: only gain changes, while the other 11 parameters remain fixed. Curves are expressed as loss change relative to their own center loss; different centers are not connected into a false global landscape.

Cell 1: individual local loss slices

Cell 1: local STRF-gain slices at Latin-hypercube parameter contexts

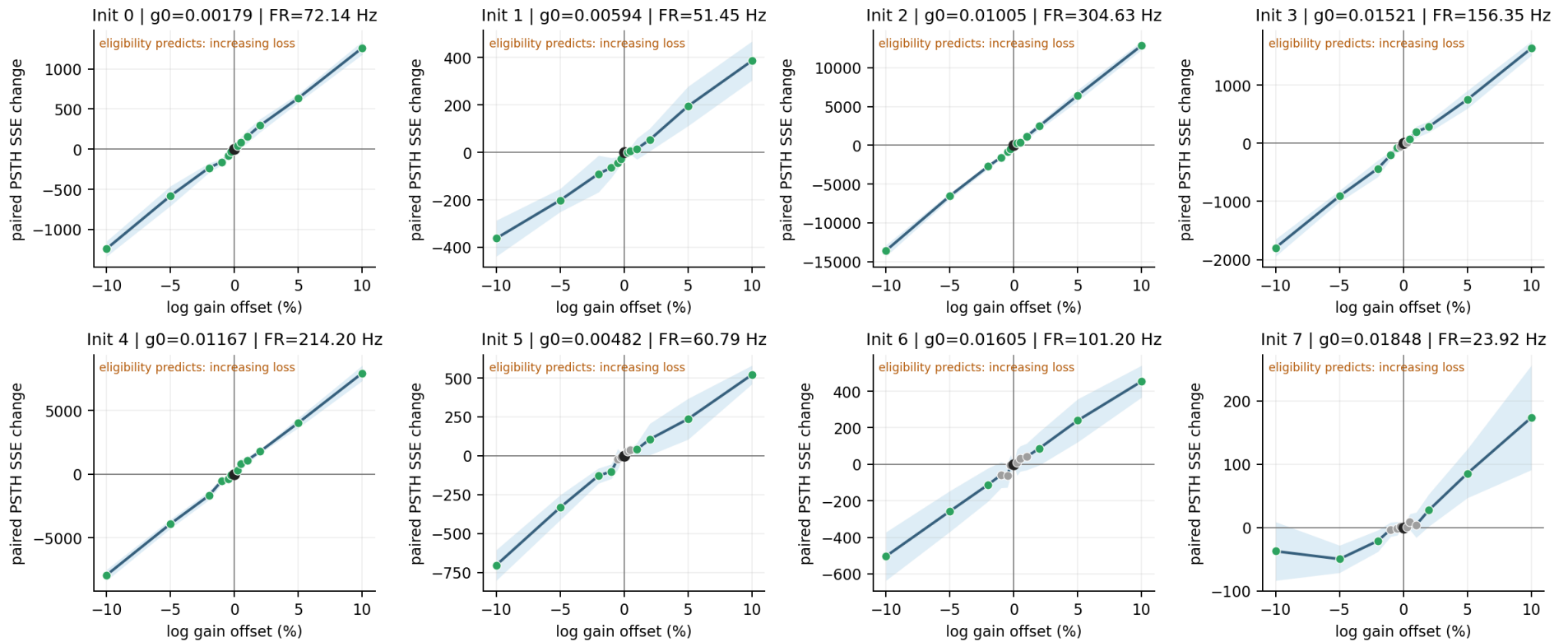


Point color is the qualification of the symmetric +/- pair at that epsilon, not an endpoint-specific test.

The shaded band is the nominal 95% seed-level interval for paired loss change. Point colors apply to the symmetric +/- pair at that epsilon. Orange text gives only the archived center direction so the actual loss curve remains visually readable.

Cell 2: individual local loss slices

Cell 2: local STRF-gain slices at Latin-hypercube parameter contexts

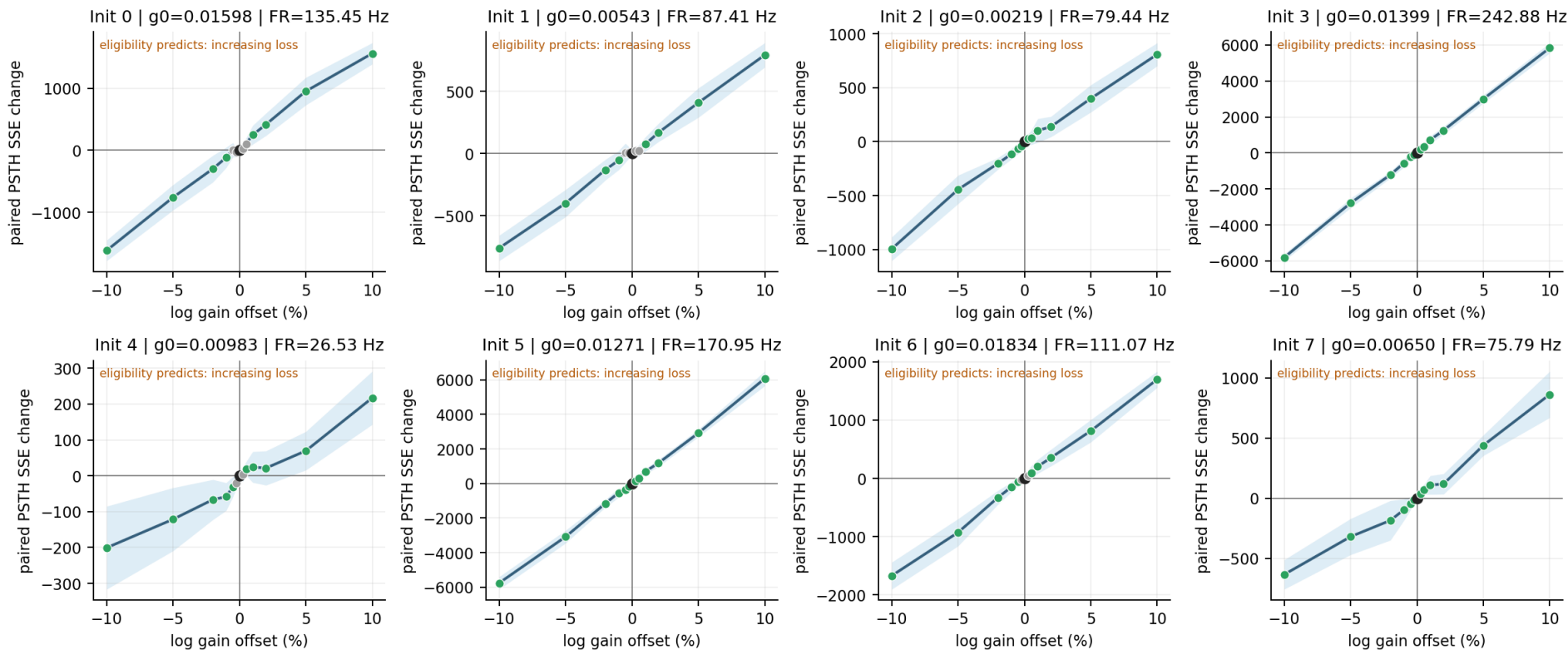


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The shaded band is the nominal 95% seed-level interval for paired loss change. Point colors apply to the symmetric +/- pair at that epsilon. Orange text gives only the archived center direction so the actual loss curve remains visually readable.

Cell 7: individual local loss slices

Cell 7: local STRF-gain slices at Latin-hypercube parameter contexts

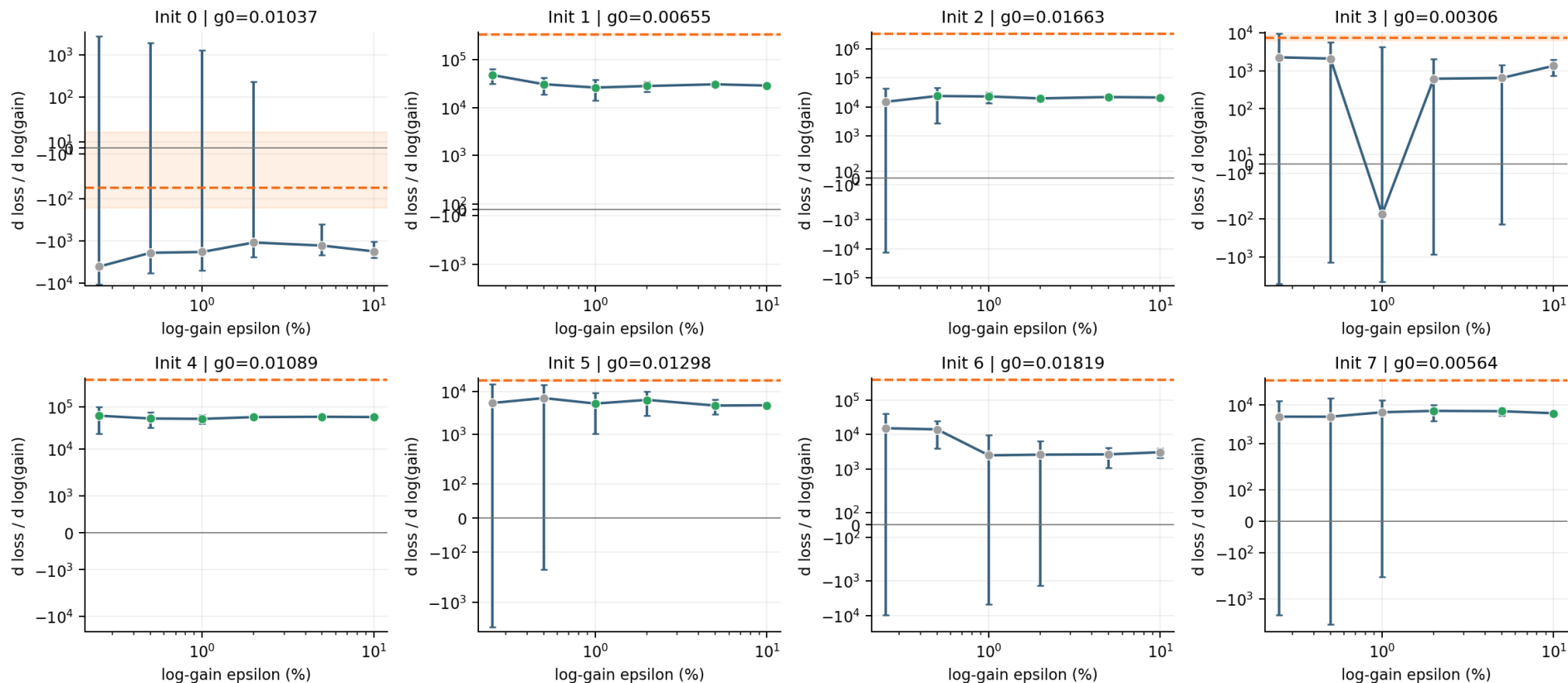


Point color is the qualification of the symmetric +/- pair at that epsilon, not an endpoint-specific test.

The shaded band is the nominal 95% seed-level interval for paired loss change. Point colors apply to the symmetric +/- pair at that epsilon. Orange text gives only the archived center direction so the actual loss curve remains visually readable.

Cell 1: finite-difference epsilon sensitivity

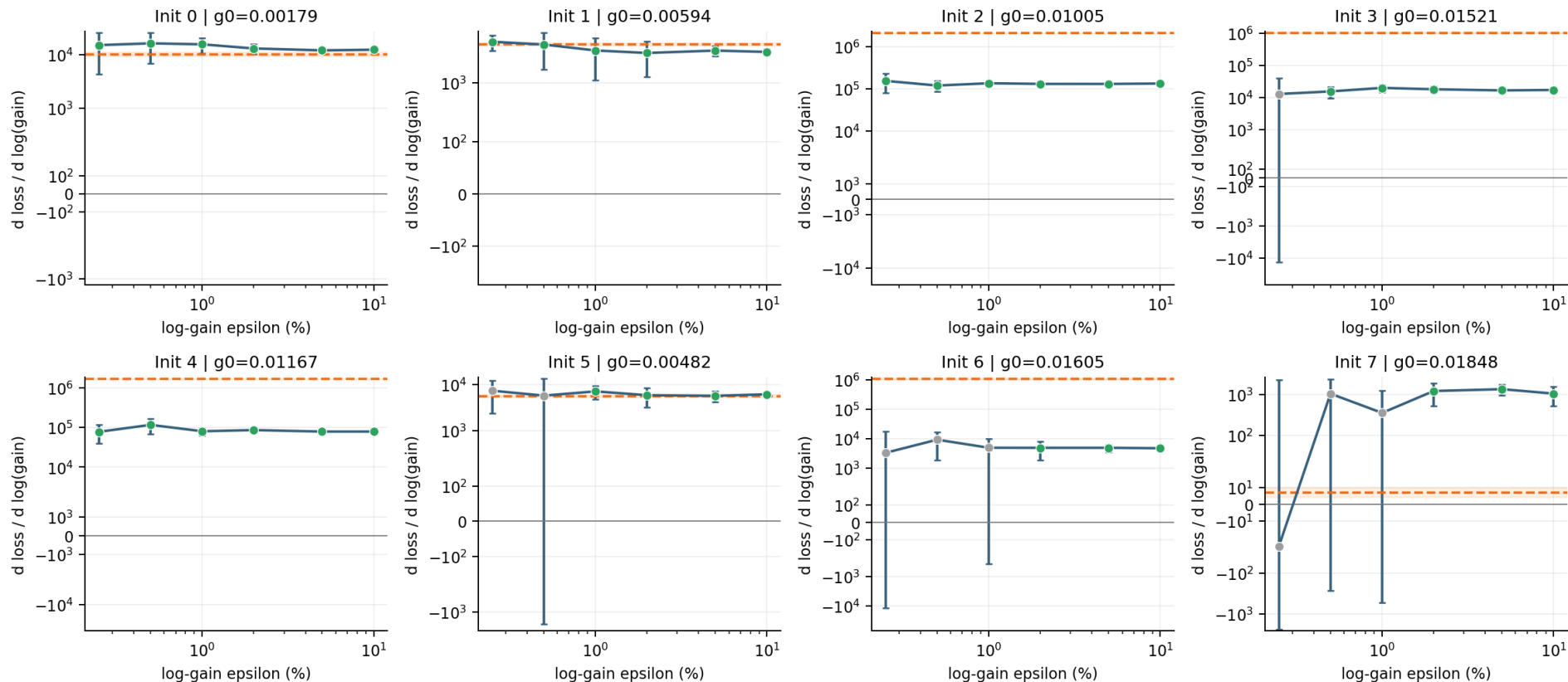
Cell 1: central-difference epsilon sensitivity (orange = archived eligibility)



Blue points and intervals are paired central secants of expected sampled loss. The orange dashed line and band show archived center eligibility. Stable numerical slopes across epsilon support a local directional conclusion, while vertical separation from orange shows magnitude miscalibration.

Cell 2: finite-difference epsilon sensitivity

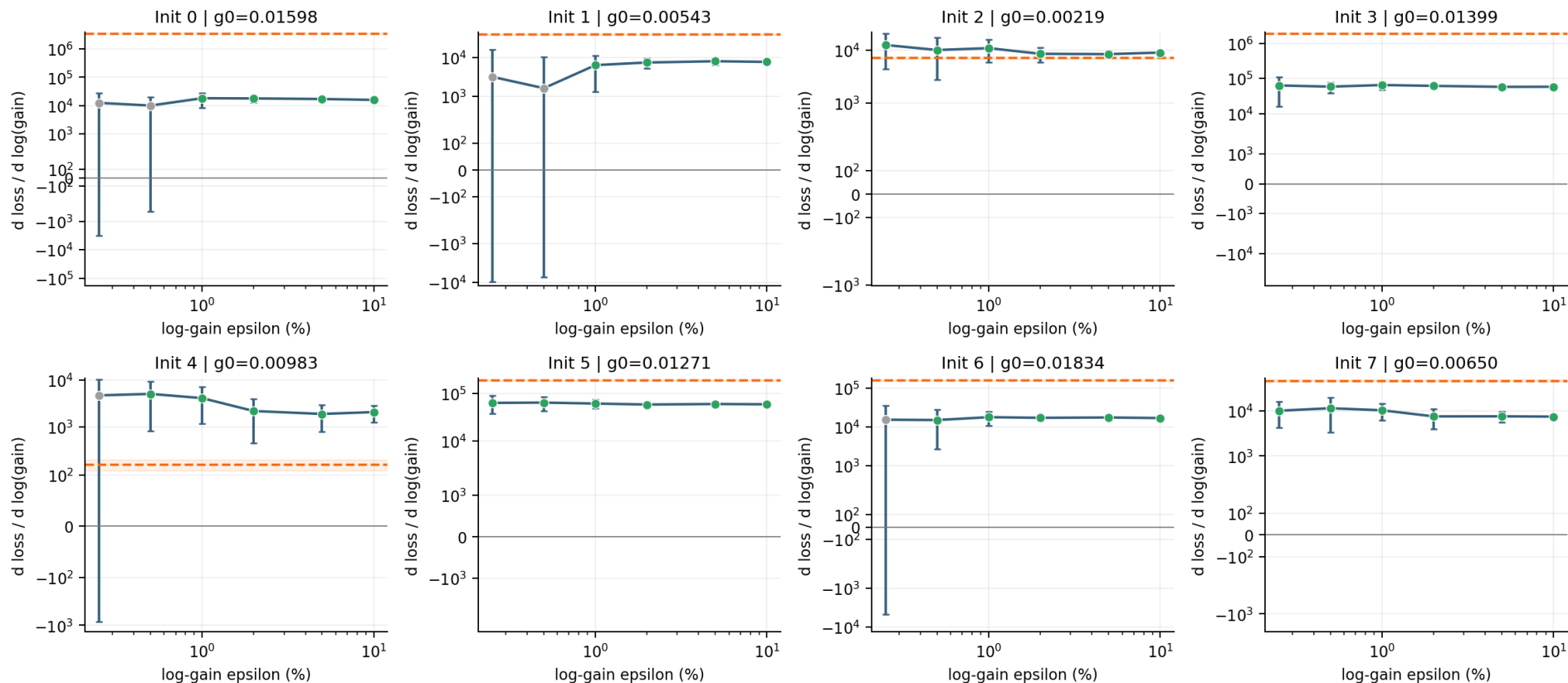
Cell 2: central-difference epsilon sensitivity (orange = archived eligibility)



Blue points and intervals are paired central secants of expected sampled loss. The orange dashed line and band show archived center eligibility. Stable numerical slopes across epsilon support a local directional conclusion, while vertical separation from orange shows magnitude miscalibration.

Cell 7: finite-difference epsilon sensitivity

Cell 7: central-difference epsilon sensitivity (orange = archived eligibility)



Blue points and intervals are paired central secants of expected sampled loss. The orange dashed line and band show archived center eligibility. Stable numerical slopes across epsilon support a local directional conclusion, while vertical separation from orange shows magnitude miscalibration.

All 24 primary comparisons

Cell	Init	g0	FR	FD dL/dlog(g)	Elig dL/dlog(g)	FD/elig	FD 95% CI	Elig 95% CI	Status
1	0	0.01037	37.2	-1078	-67.25	16.03	[-2.39e+03, 234]	[-162, 27.6]	gray
1	1	0.006546	160.6	2.808e+04	3.4027e+05	0.08253	[2.11e+04, 3.5e+04]	[3.284e+05, 3.521e+05]	green
1	2	0.01663	218.7	1.977e+04	3.3997e+06	0.005814	[1.48e+04, 2.47e+04]	[3.322e+06, 3.477e+06]	green
1	3	0.003062	61.2	618.8	7592	0.08151	[-860, 2.1e+03]	[6.37e+03, 8.81e+03]	gray
1	4	0.01089	252.9	5.768e+04	4.5482e+05	0.1268	[5.26e+04, 6.28e+04]	[4.432e+05, 4.665e+05]	green
1	5	0.01298	91.3	6309	1.824e+04	0.346	[2.68e+03, 9.94e+03]	[1.76e+04, 1.89e+04]	green
1	6	0.01819	85.1	2584	3.9252e+05	0.006584	[-1.39e+03, 6.56e+03]	[3.807e+05, 4.043e+05]	gray
1	7	0.005645	94.8	6944	4.277e+04	0.1624	[3.84e+03, 1e+04]	[4.14e+04, 4.41e+04]	green
2	0	0.001787	72.1	1.319e+04	1.035e+04	1.274	[1.03e+04, 1.61e+04]	[9.56e+03, 1.11e+04]	green
2	1	0.005941	51.5	3581	5203	0.6883	[1.29e+03, 5.88e+03]	[4.92e+03, 5.49e+03]	green
2	2	0.01005	304.6	1.2981e+05	2.1260e+06	0.06106	[1.167e+05, 1.429e+05]	[2.095e+06, 2.157e+06]	green
2	3	0.01521	156.3	1.802e+04	1.0765e+06	0.01674	[1.37e+04, 2.23e+04]	[1.060e+06, 1.093e+06]	green
2	4	0.01167	214.2	8.676e+04	1.7350e+06	0.05001	[7.84e+04, 9.51e+04]	[1.701e+06, 1.769e+06]	green
2	5	0.004819	60.8	5853	5683	1.03	[3.22e+03, 8.48e+03]	[5.31e+03, 6.06e+03]	green
2	6	0.01605	101.2	4956	1.1028e+06	0.004494	[1.88e+03, 8.03e+03]	[1.059e+06, 1.147e+06]	green
2	7	0.01848	23.9	1219	7.123	171.1	[520, 1.92e+03]	[4.21, 10]	green
7	0	0.01598	135.5	1.787e+04	3.6005e+06	0.004965	[1.29e+04, 2.28e+04]	[3.507e+06, 3.694e+06]	green
7	1	0.005434	87.4	7491	4.063e+04	0.1844	[5.33e+03, 9.65e+03]	[3.89e+04, 4.24e+04]	green
7	2	0.00219	79.4	8544	7165	1.192	[5.9e+03, 1.12e+04]	[6.89e+03, 7.44e+03]	green
7	3	0.01399	242.9	6.164e+04	1.9263e+06	0.032	[5.46e+04, 6.86e+04]	[1.897e+06, 1.956e+06]	green
7	4	0.009829	26.5	2200	164.2	13.4	[460, 3.94e+03]	[122, 207]	green
7	5	0.01271	170.9	5.845e+04	1.9124e+05	0.3057	[5.21e+04, 6.48e+04]	[1.863e+05, 1.962e+05]	green
7	6	0.01834	111.1	1.711e+04	1.5861e+05	0.1079	[1.43e+04, 1.99e+04]	[1.518e+05, 1.654e+05]	green
7	7	0.006496	75.8	7562	4.726e+04	0.16	[4e+03, 1.11e+04]	[4.45e+04, 5e+04]	green

FR is the center firing rate in Hz. Gradients use the log-gain coordinate so signs and units are directly comparable.

Conclusions and limitations

1. **The concern about the first report was valid.** Its broad neighboring points measured long secants and created many inconclusive classifications. Small paired perturbations provide much stronger local evidence for STRF gain.
2. **Direction is robust in this pilot.** At the predeclared 2% scale, 21/24 contexts were qualified green, none were qualified red, and all 24 mean signs agreed. Results were similar from 1% through 10%.
3. **Magnitude is not validated.** Numerical FD values are generally smaller than archived eligibility values and the calibration ratio varies substantially by context. This can arise from surrogate-rule scaling, missing normalization, omitted paths, or hard-event/expected-loss differences; this experiment alone does not identify which.
4. **This is an initialization-distribution test.** Eight Latin-hypercube batches sparsely cover a 12-dimensional space, and their firing rates are often high. The test should next be repeated around fitted checkpoints and biologically relevant firing-rate regimes.
5. **The uncertainty analysis is exploratory.** Eight seed-level observations give low-power nominal t intervals for a discrete, potentially heavy-tailed difference. Epsilon tests are correlated, contexts share the seed's spontaneous input, and no multiplicity correction is applied.
6. **This does not test a frozen-event state derivative.** A separate voltage/rate sensitivity finite difference with event sequences held fixed is still the cleanest way to isolate chain-rule implementation from the stochastic spike objective.

Bottom line: the archived STRF-gain eligibility appears to point downhill/uphill correctly across the sampled local contexts, but the analysis does not support calling it the exact gradient of expected PSTH SSE. The next decisive experiment is the same paired epsilon ladder around trained parameter states, followed by an explicit investigation of the magnitude scale discrepancy.